

Customer Description

The existing **Configurable Edge Pico Platform** provides a modular edge-computing framework capable of supporting configurable services, sensor integration, edge analytics, and deployable AI workloads on the Raspberry Pi Pico platform.

We want to develop an **ESP32-S3 implementation of the Edge Pico Platform**, preserving the core architecture and functionality while leveraging the enhanced capabilities of the ESP32-S3.

Students should assume that the original platform architecture already exists and serves as the reference implementation. The objective is to reproduce the major platform capabilities on the ESP32-S3 and demonstrate equivalent or improved functionality.

The platform should continue to support:

- Configurable edge applications
- Modular software components
- Sensor and actuator integration
- Edge data processing
- AI inference workloads
- Network-connected edge services

Beyond feature parity, students are encouraged to identify opportunities to exploit ESP32-S3-specific capabilities to improve system performance, scalability, and usability.

Particular emphasis should be placed on AI and machine learning workloads. The ESP32-S3 provides significantly greater computational resources, memory capacity, wireless connectivity, and AI-oriented instruction support than the original platform. Students should investigate how these capabilities can be utilized to improve inference performance, increase model complexity, reduce latency, or enhance energy efficiency.

The completed platform should demonstrate that applications originally developed for Edge Pico can operate on the ESP32-S3 while benefiting from the strengths of the new hardware platform.

Motivation

This project builds upon the previous **Configurable Edge Pico Platform**, which focused on creating a configurable and reusable edge-computing architecture.

While the original project emphasized modularity and extensibility, this project focuses on implementation maturity and performance.

Many embedded platforms begin life as proof-of-concept systems before being migrated to higher-performance hardware that better supports networking, AI workloads, and production deployment requirements.

The ESP32-S3 is an attractive target platform because it provides:

- Integrated Wi-Fi
- Bluetooth Low Energy (BLE)
- Enhanced processing capability
- Increased memory resources
- Vector instruction support for AI workloads
- Rich peripheral support

Students must evaluate how these additional resources can be used to enhance platform capabilities while maintaining compatibility with the original Edge Pico design philosophy.

Short Product Vision Statement

Develop an **ESP32-S3 implementation of the Configurable Edge Pico Platform** that reproduces the functionality of the original system while improving performance, connectivity, and AI inference capabilities.

Requirements Students Can Infer

Area	Questions students should resolve
Platform Migration	Which Edge Pico features must be reproduced?
Software Architecture	How can the original component framework be implemented on ESP32-S3?
Networking Integration	How can Wi-Fi and BLE be incorporated into the platform?
Sensor Integration	How should existing drivers and services be supported?
Configuration Engine	Can existing deployment and configuration approaches be maintained?

AI Acceleration	How can ESP32-S3 resources improve ML inference performance?
Memory Optimization	How can larger memory resources be exploited effectively?
Component Reuse	Which Edge Pico modules can be reused directly?
Performance Profiling	How should speed, memory usage, and latency be measured?
Resource Utilization	How efficiently are CPU, memory, and wireless resources used?
Verification	How can feature equivalence with the original platform be demonstrated?
Future Expansion	What new capabilities become possible on ESP32-S3?

Example Validation Scenarios

Scenario 1: Feature Parity Demonstration

An application that runs on the original Edge Pico platform is deployed on the ESP32-S3.

Expected outcome:

- Equivalent functionality.
- Identical configuration workflow.
- Minimal application-level modifications.

Scenario 2: Edge AI Performance Evaluation

An AI model executed on Edge Pico is deployed on ESP32-S3.

Expected outcome:

- Reduced inference latency.
- Improved throughput.
- Support for larger or more sophisticated models.

Scenario 3: Connected Edge Device

The platform utilizes ESP32-S3 wireless capabilities.

Expected outcome:

- Remote configuration and monitoring.
- Cloud or local network integration.
- Reliable wireless operation.

Scenario 4: Multi-Service Deployment

Multiple platform services operate simultaneously.

Expected outcome:

- Stable execution.
- Predictable performance.
- Efficient resource utilization.